# NIKHIL KUMAR TIWARI

**FULL-STACK DEVELOPER • COMPUTER SCIENCE (AI/ML) STUDENT**
+91 7986156603 | nikhilkumar.0112358@gmail.com | LinkedIn |Portfolio | Github

## PROFESSIONAL SUMMARY

Computer Science (AI/ML) student and Full-Stack Developer experienced in building and deploying web applications using MERN and Next.js. Hands-on experience with Retrieval-Augmented Generation (RAG), vector embeddings, and LLM-powered applications. Strong foundation in backend development, databases, caching, and system design fundamentals, with a focus on building practical, scalable software solutions.

## TECHNICAL SKILLS

- Languages: C++, Python, JavaScript, TypeScript
- Frontend: React.js, Next.js, Tailwind CSS
- Backend: Node.js, Express.js, REST APIs, Authentication
- AI / GenAI: LLMs, RAG, Vector Embeddings, LangChain
- Databases & Caching: PostgreSQL, MongoDB, Redis, Prisma, Pinecone
- Tools & Concepts: Git, GitHub, Docker, Socket.IO, DSA, OOP, System Design Fundamentals

## EDUCATION

**Bachelor of Technology, Computer Science (AI/ML)** — Expected 08/2029
VITBhopalUniversity,Bhopal | CGPA:8.37/10
**Intermediate (Class XII)** — 08/2024
ArmyPublic School,Jodhpur | FinalGrade:83.4%

## PROJECTS

**LeetCode Clone** | *Next.js, PostgreSQL, Prisma, Docker, Judge0, Better Auth*
*GitHub*

- Engineered a full-stack competitive coding platform enabling multi-language code compilation, sandboxed execution, and instant verdict generation via Judge0 integration.
- Implemented secure authentication and role-based access control (RBAC) with Better Auth, enforcing protected routes and session management across admin and user roles.
- Built problem management, test-case validation, and submission-history modules; containerized the full backend and execution environment with Docker for reproducible deployments.
- Modeled a normalized relational schema in PostgreSQL with Prisma ORM to support problems, users, and submissions with type-safe queries.

**VisitorPass Management System** | *MongoDB, Express.js, React.js, Node.js*
*GitHub*

- Built an end-to-end visitor management system automating request submission, multi-step approval workflows, and QR-code-based entry verification.
- Designed and developed RESTful APIs with role-based workflows to control visitor access across admin, host, and visitor roles.
- Integrated automated email notifications to streamline approval communication and remove manual coordination overhead.

**Full-StackChat Application** | *React.js, Node.js, Express.js, MongoDB, Socket.IO*
*GitHub*

- Developed a real-time messaging platform using Socket.IO for bidirectional, low-latency communication with live online/offline presence tracking and JWT-based authentication.
- Built image sharing and profile upload features with persistent MongoDB storage, delivering a complete real-time social messaging experience.

## CERTIFICATIONS

- Tutedude — MERN Stack Development Course
- Complete React and Next.js Course with AI-Powered Projects